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International Mathematics Assessments for Schools

2024 / 2025 JUNIOR DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2024 / 2025 JUNIOR DIVISION FIRST ROUND PAPER

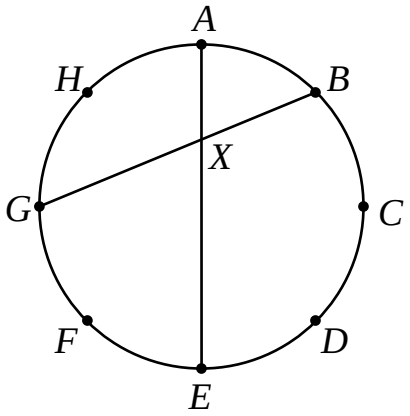
Questions 1-10, 3 marks each

1. There is a 48-story building which has 24 windows per floor. A cleaner starts his work on the 48th floor and needs to clean all the windows on each floor before moving down to the floor below. On which floor will the cleaner be after he has cleaned exactly 177 windows?

   （A）39　　（B）40　　（C）41　　（D）42　　（E）43

---

2. In the diagram below, points *A*, *B*, *C*, *D*, *E*, *F*, *G* and *H* divide the circumference into 8 equal parts. If *X* is the intersection of *AE* and *BG*, what is the measure, in degrees, of $\angle GXE$?



   （A）45°　　（B）52.5°　　（C）60°　　（D）67.5°　　（E）75°

---

3. Let *p* be a prime number and *q* be a positive integer such that $10p+4q$ and $28p+4q$ are both perfect squares. What is the value of $p+q$?

   （A）9　　（B）10　　（C）11　　（D）12　　（E）13

---

4. A convex polygon has *n* interior angles. If its interior angles are consecutive odd integers, in degrees, and the smallest angle among them is 139°, what is the value of *n*?

   （A）8　　（B）10　　（C）12　　（D）16　　（E）18

---

5. What are the last three digits of 11^{2025}?

   （A）121　　（B）251　　（C）331　　（D）561　　（E）641

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6. A mountain cable car company charges each passenger \$10 for each station that they visit (excluding the station where they boarded the cable car) during the cable car ride. For example, if a passenger gets on a cable car at the 2nd station and then gets off at the 5th station, his total fare would be \$30.
All cable cars are empty when they reach the 1st station which is located at the bottom of the mountain and all passengers must get off at the final station which is located on the top of the mountain. The company has to record how many passengers get on and off these stations throughout the day. At each station, the notation (*a*, *b*) means that *a* one-way passengers get on the cable cars at this station and *b* one-way passengers get out of this station. For example, if the company records (35, 20) for the second station and (45, 15) for the third station, this means that 35 passengers get on and 20 passengers get off at the second station, and then, at the third station, 45 more passengers get on and 15 passengers get off.
One day, the records for each of the stations are as follows:

1st station	2nd station	3rd station	4th station	5th station	6th station	Final station
(50, 0)	(25, 10)	(33, 22)	(44, 25)	(16, 23)	(10, 38)	(0, 60)

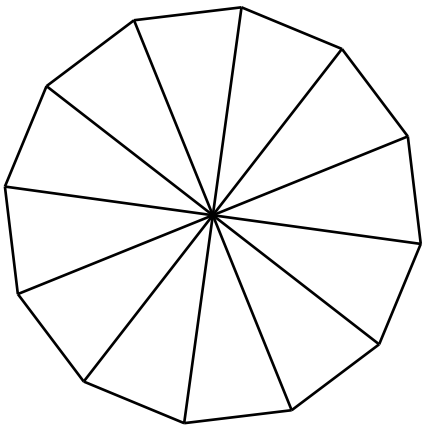
What is the total amount that all the passengers pay?
（A）\$4340　（B）\$4330　（C）\$4320　（D）\$4310　（E）\$4300

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7. Six adults and three children went camping. The camping program includes hiking up into the mountains. Any group interested in going hiking must have at least two adults. How many different hiking groups could be formed out of the nine people?
（A）296　（B）336　（C）408　（D）456　（E）512

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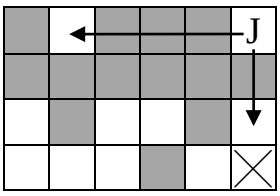
8. Amir has 12 pieces of isosceles triangles, where the length of the common side is 18 cm. He arranges them, without overlapping, to form a regular dodecagon (12 sides) as shown in the diagram below. What is the area, in cm^2, of this dodecagon?



（A）81　（B）162　（C）324　（D）486　（E）972

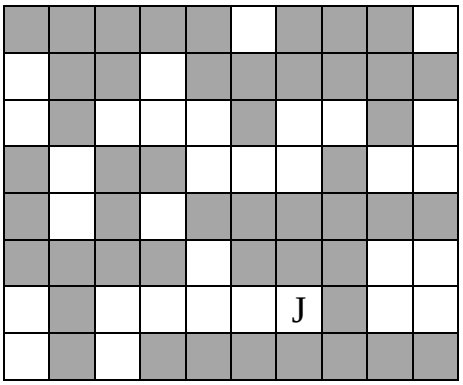
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9. In a computer game, a maze is illustrated as a grid, where some cells are painted in grey. The grey cells indicate where traps have been placed and the white cells indicate safe cells. A jumping man named Jay is standing on a white cell, which is indicated as "J". Jay may jump from one white cell to another white cell. However, he may only jump horizontally or vertically. He can jump over a sequence of grey cells, but he cannot jump over another white cell (and he cannot land on a grey cell).



For example, as shown in the diagram on the right, if Jay stands on the white cell in the top right corner, he may jump four cells to the left or he may jump two cells down, but he cannot jump three cells down. Also, if he wants to get to any other white cell, he would need to make additional jumps.

In the following maze, Jay begins on the white cell marked with "J". What is the minimum number of jumps he needs to do to be sure that any other white cell can be reached when jumping from this starting position?



(A) 6 (B) 7 (C) 8 (D) 9 (E) 10

10. In both diagrams below, neighbouring regions mean that they share a common segment. If two regions just share a common point without a common segment, then these two regions are not neighbouring regions. For example, in diagram B, the regions 1 and 5 are not neighbouring regions. In both diagrams below, Alex wants to paint the regions that have the same number using same colour such that no neighbouring regions are painted in the same colour. What is the minimum number of colours that Alex needs?

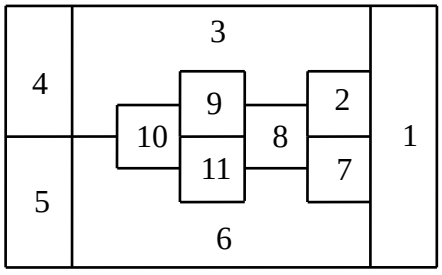


Diagram A

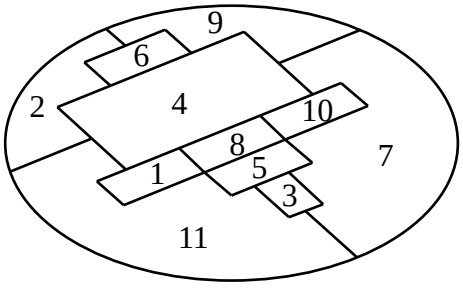


Diagram B

(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

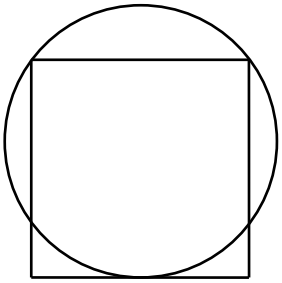
Questions 11-20, 4 marks each

11. A class of seven students is to be divided into two groups, where the first group consists of four students, while the second group contains three students. Each group must choose one or two students to join a competition. In how many ways can they form groups to join the competition?

（A）2100　（B）2160　（C）1050　（D）560　（E）315

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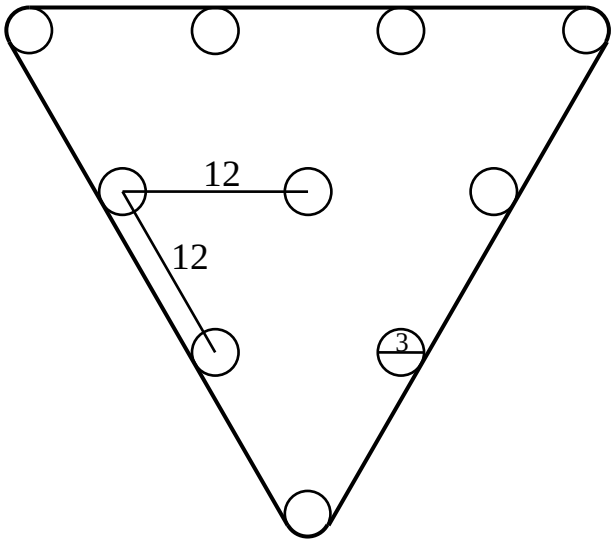
12. In the diagram below, a square and a circle are tangent at one point and they intersect at four other points. If the side length of the square is 4 cm, what is the diameter, in cm, of the circle?



（A）2.5　（B）3　（C）4　（D）5　（E）6

---

13. In Candlepin Bowling, ten cylindrical pins, each having a diameter of 3 inches, are arranged in a triangular layout, with the distance between the centres of the adjacent pins being 12 inches, as shown in the diagram below. If we use a rope having the shortest possible length to enclose the ten pins without moving them, as shown by the thick curve, the enclosed area is $a+b\sqrt{3}+\frac{c}{4}\pi$ squared inches, where a, b and c are integers. What is the value of $a+b+c$?



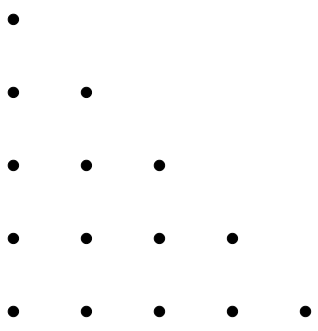
（A）252　（B）324　（C）333　（D）495　（E）504

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14. How many positive integers n less than 2025 have the property that $n = a + b + c$, where a, b and c are three distinct factors of n?
（A）225　（B）333　（C）337　（D）450　（E）900

15. Andy, Benny and Charlie will run on three straight parallel tracks a, b and c (track b is between tracks a and c). At the start of the race, all three runners are on a line perpendicular to the three tracks. The distance between tracks a and b is twice the distance between tracks b and c. During the whole race, the three runners will all run on a straight line. If the speeds of Andy and Charlie are 8 km/h and 20 km/h, respectively, what is the speed, in km/h, of Benny?
（A）4　（B）8　（C）12　（D）16　（E）20

16. We randomly select three points from the following grid of 15 points as shown in the diagram below, where each of the points have the same chance of being chosen. If $\frac{p}{q}$ (written as an irreducible fraction) is the probability that with these three points one can form a non-degenerate triangle, what is the value of $p + q$?
(Note: A non-degenerate triangle is a triangle with a non-zero area.)

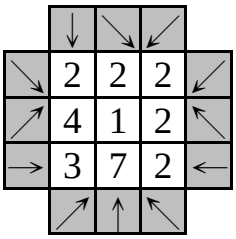


（A）865　（B）862　（C）172　（D）123　（E）25

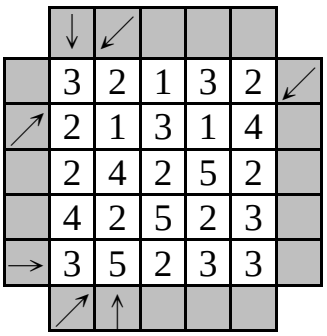
17. Tom is now x years old and the ages of his five cousins are 28, 33, 37, 47 and 52. What is the sum of all possible values of x such that the average age of all six people (Tom and his five cousins) is equal to their median age?
（A）222　（B）111　（C）98　（D）68　（E）56

18. A bowling alley has x ten-pound bowling balls, y twelve-pound bowling balls and z fourteen-pound bowling balls, where there is at least one bowling ball of each weight. It is known that the total weight of all the bowling balls is 200 pounds, y is a multiple of 3 and $x > z$. What is the sum of all possible values of x?
（A）15　（B）18　（C）20　（D）33　（E）38

19. Some shaded squares are located around a square grid must be filled with one arrow each, drawn in a way that the arrow points vertically (↑ and ↓), horizontally (→ and ←) or diagonally (↘, ↙, ↖ and ↗) along the grid. Each arrow must point to at least one unshaded square. The number of arrows pointing to each unshaded square in the grid equals the number shown in that square. The following diagram shows one example:



Ben has to completely fill in the arrows in the grid below, where some arrows were already pre-filled. After completing the task, what is the total number of arrows ↖ and ↘ arrows that Ben used?



（A）6　　（B）7　　（C）8　　（D）9　　（E）10

20. Annie wrote eleven consecutive two-digit numbers in a row:

$$a,\ a+1,\ a+2,\ ...,\ a+10.$$

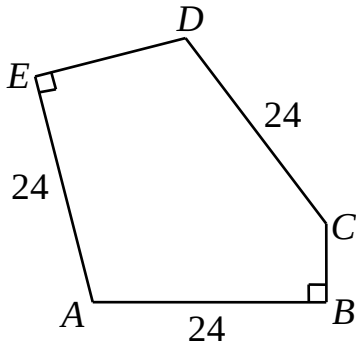
When she erased three numbers, the average of the original numbers is the same as the average of the remaining numbers. If the three numbers erased by Annie are the x-th, y-th and z-th numbers from left to right, what is the number of all possible ordered triples (x, y, z)?

（A）14　　（B）13　　（C）12　　（D）11　　（E）10

Questions 21-25, 6 marks each

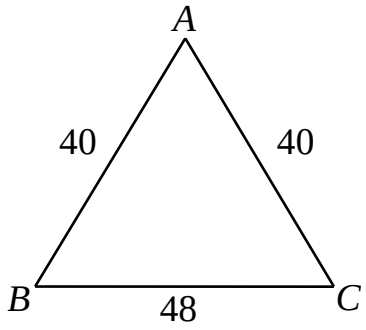
21. A bag only contains some number of black and white balls. The ratio of black to white balls is 1:2. If two balls are randomly drawn from the bag at the same time, the probability that the two balls have different colours is $\frac{64}{143}$. How many balls are there in the bag in total?

22. Let $ABCDE$ be a pentagon such that $AB = AE = CD = 24\,\text{cm}$, $BC + DE = 24\,\text{cm}$ and $\angle ABC = \angle DEA = 90°$. What is the area, in cm^2, of the pentagon?



23. What is the least possible integer n such that $n! = 1 \times 2 \times 3 \times ... \times n$ would have at least 200 trailing zeros?

24. Let ABC be an isosceles triangle with $AB = AC = 40$ cm and $BC = 48$ cm. Let the radii of the circumscribed and inscribed circles of triangle ABC be R and r, respectively. What is value, in cm, of $R - r$?



25. Let a_1, a_2, …, a_n be distinct positive integers such that

$$\frac{a_1 + 4}{a_1} \times \frac{a_2 + 4}{a_2} \times ... \times \frac{a_n + 4}{a_n} = 2025.$$

What is the least possible value of n?

* * *